Evidence of a Floquet Phase in a Photonic System

Ernesto P. Raposo,[1] Iván R. R. González,[1] A. M. S. Macêdo,[1] Bismarck C. Lima,[2] Raman Kashyap,[3] Leonardo de S. Menezes,[2] and Anderson S. L. Gomes[2]

[1]*Laboratório de Física Teórica e Computacional, Departamento de Física, Universidade Federal de Pernambuco, 50670-901 Recife, Pernambuco, Brazil*

[2]*Departamento de Física, Universidade Federal de Pernambuco, 50670-901 Recife, Pernambuco, Brazil*

[3]*Fabulas Laboratory, Department of Engineering Physics, Department of Electrical Engineering, Polytechnique Montreal, Montreal, H3C 3A7, Quebec, Canada*

(Received 4 October 2018; revised manuscript received 26 January 2019; published 12 April 2019)

The ground breaking extension of the key concept of phase structure to nonequilibrium regimes was only recently achieved in Floquet systems, characterized by a time-dependent quantum Hamiltonian with a periodic driving source. However, despite the theoretical advances, only very few systems are known to display experimental Floquet phases, not one of them employing a laser emission-based mechanism. Here we report the first experimental observation of a Floquet phase in a photonic system, a disordered fiber laser with spatial eigenmode localization. We apply a periodically oscillating cw pumping source that drives the random couplings of the Floquet Hamiltonian. A photonic Floquet spin-glass phase is demonstrated in the random-lasing regime by extensive measurements of the Parisi overlap parameter and asymmetry properties of its distribution. In contrast, in the fluorescent regime below threshold, the absence of mode localization prevents the stabilization of a Floquet phase. Our results are nicely described by theoretical arguments.

DOI: 10.1103/PhysRevLett.122.143903

Thermalization and phase structure of interacting systems are central principles of equilibrium (either classical or quantum) physics. The fundamental quest for the extension of the concept of phase structure to nonequilibrium systems has ignited a recent burst of interest [1–11] in quantum disordered many-body interacting Floquet systems, described by a periodically driven Hamiltonian.

On the one hand, a trivial unstructured "infinite temperature" state generally emerges in nonequilibrium interacting systems in contact with a driving energy source [5]. Remarkably, however, the interplay of disorder and localization ingredients can stabilize a nontrivial structure of dynamical Floquet phases in out-of-equilibrium systems. A suitable amount of disorder can lead to spatially localized eigenmodes that couple weakly with faraway modes. In this case, the energy transfer among modes is hampered and the energy absorption from the driving source saturates, preventing the route to the infinite temperature state, a phenomenon known as many-body localization.

On the theoretical side, a number of Floquet systems with nontrivial phase structure have been recently reported, e.g., in quantum disordered spin Hamiltonians with periodic couplings displaying a Floquet spin-glass phase [1–7]. However, the experimental scenario advances in a much slower pace, with only quite few systems presenting experimental Floquet phases, most of them regarding discrete time crystals [9–11] and none employing a laser emission-based photonic mechanism.

In this work we report the first experimental evidence of a Floquet spin-glass phase in a photonic system, a quasi-one-dimensional erbium-based multimode random fiber laser (RFL). Random lasers and RFLs have been recently employed as successful platforms to access nontrivial complex phases, such as equilibrium spin glasses [12–18] and turbulentlike regimes [19–22], as theoretically justified in Refs. [20–24]. The interplay of disorder, localization, and nonlinearity in the multimode RFL system studied in this work is essential to stabilize the observed photonic spin-glass Floquet phase. Disorder was introduced by inscribing a remarkable amount ($\sim 10^3$) of specially designed stationary random Bragg gratings in the fiber core [25]. Using the speckle contrast technique, the RFL was shown [17] to emit a large number (>200) of longitudinal modes. The strong quenched disorder is also responsible for the spatial localization of modes above the RFL threshold, as demonstrated in Ref. [25].

The RFL is pumped by a periodic continuous-wave (cw) source that drives directly the quadratic and quartic couplings in the Floquet Hamiltonian. The system is submitted to a large number of periodic oscillations of the drive and the correlation statistics of a rather extensive number (150 000) of RFL emission spectra above threshold is analyzed. Specifically, a time-dependent correlator is measured in the form of a Parisi overlap parameter that quantifies the correlation between mode configurations in RFL spectra separated in time. In the strongly nonlinear disordered regime above threshold, in which the spatial mode localization is effective, we observe that the distribution of overlap values for a given separation time presents a bimodal pattern, with two side maxima typical of spin-glass phases. However, in contrast with equilibrium spin glasses, the intensity of the maxima can be highly asymmetrical, depending on the separation time, and evolves dynamically with the same period of the drive. The observed periodically driven Floquet spin-glass phase thus leads to a dynamics of the RFL spectra that alternates periodically between mostly correlated and mostly anti-correlated mode configurations. In contrast, when mode localization is suppressed, as in the high-loss fluorescent regime below threshold, we observe that no Floquet phase can establish.

Theoretical background.—The erbium-based multimode RFL system can be modeled by a system-bath Hamiltonian in the form $\mathcal{H} = \mathcal{H}_S + \mathcal{H}_B + \mathcal{H}_{SB}$, where the first two terms describe, respectively, the quantized electromagnetic field inside and outside the RFL, while $\mathcal{H}_{SB}$ accounts for the system-bath interaction. By considering $\mathcal{H}_B$ as a large reservoir of quantum oscillators, the Feshbach projection technique separates [26,27] the contributions to the field from the regions inside and outside the RFL, leading [23,27] in the rotating-wave approximation to the effective Hamiltonian

$$\mathcal{H}_S = -\frac{1}{2}\sum_{k_1 k_2} g^{(2)}_{k_1 k_2} a^\dagger_{k_1} a_{k_2} - \frac{1}{4!}\sum_{k_1 k_2 k_3 k_4} g^{(4)}_{k_1 k_2 k_3 k_4} a^\dagger_{k_1} a_{k_2} a^\dagger_{k_3} a_{k_4}, \tag{1}$$

where $a^\dagger_k$ and a_k are creation and annihilation operators assigned to the field eigenmodes indexed by k. The relevant fact to the purpose of this work is that the quadratic $g^{(2)}$ and quartic $g^{(4)}$ couplings in Eq. (1) depend directly on the pump intensity S in a form proportional to $S\int d\vec{r}\rho(\vec{r})f^*_{k_1}(\vec{r})f_{k_2}(\vec{r})$ and $S\int d\vec{r}\rho(\vec{r})f^*_{k_1}(\vec{r})f_{k_2}(\vec{r})f^*_{k_3}(\vec{r})f_{k_4}(\vec{r})$, respectively, where $f_k(\vec{r})$ denotes the atom-field coupling and $\rho(\vec{r})$ is the density of erbium ions in the fiber that provide the gain generating the feedback for the random laser emission above the RFL threshold. The quartic coupling $g^{(4)}$ is the main source of nonlinearity in the RFL, associated with the nonlinear susceptibility $\chi^{(3)}$ due to the single-photon induced nonlinear absorption. Moreover, the highly disordered character of the couplings $g^{(p)}$, both quadratic ($p = 2$) and quartic ($p = 4$), arises from the random positions of the erbium ions and random spatial profile of the RFL refractive index related to the multiple random Bragg gratings inscribed in the fiber.

For a fixed (i.e., time-independent) pump intensity S, the Hamiltonian (1) is similar to that of a quantum p-spin spherical model [28], with pth order random couplings between spins and spherical constraint on the sum of squared spin operators (in the photonic context, the analogous constraint corresponds to fix the total optical power [23]). Remarkably, when spin couplings with Gaussian distribution are considered with $p > 2$, an *equilibrium* spin-glass phase emerges [28] under Parisi's replica-symmetry-breaking (RSB) formalism. Moreover, in the semiclassical context a Hamiltonian similar to Eq. (1), with $a^\dagger_k$ and a_k replaced by their expectation values, also yields a photonic equilibrium RSB spin-glass phase in multimode random laser systems in closed and open cavities, for sufficiently strong disorder in the $p = 2$ and $p = 4$ Gaussian couplings [23]. A magnetic-to-photonic analogy is evidenced from these works, in which the lasing mode k plays the role of the spin at site i, $a_k \leftrightarrow S_i$. Also, the pump intensity is analogous [23] to the inverse temperature in the magnetic system.

On the experimental side, an *equilibrium* photonic spin-glass phase with RSB was first demonstrated in a solid-state random laser [12] (see also Refs. [13–18]). In principle, the spin-glass order parameter should be inferred from the Parisi overlap parameter $Q_{\alpha\beta} \propto (1/N)\sum_k a_{k,\alpha} a^\dagger_{k,\beta}$, with N as the number of modes and the indexes α and β labeling identically prepared (replica) mode configurations associated with the emission spectra. However, as the experimental access to the phases of the operators expectation values is unfeasible in random lasers [24], in practice overlaps between intensity fluctuations determined from $a^\dagger_{k,\alpha} a_{k,\alpha}$ and $a^\dagger_{k,\beta} a_{k,\beta}$ are measured through the correlation parameter [12]

$$q_{\alpha\beta} = \frac{\sum_k \Delta_\alpha(k)\Delta_\beta(k)}{\sqrt{\sum_k \Delta^2_\alpha(k)}\sqrt{\sum_k \Delta^2_\beta(k)}}, \tag{2}$$

with intensity fluctuations $\Delta_\alpha(k) = I_\alpha(k) - \bar{I}(k)$ and $\bar{I}(k) = \sum_\alpha I_\alpha(k)/N_s$ representing the average intensity over N_s emission spectra. Indeed, it was shown in Ref. [24] that $q_{\alpha\beta} \propto Q^2_{\alpha\beta}$ in the spin-glass phase.

In the equilibrium photonic spin-glass state above threshold, the coherent oscillation of the correlated modes is frustrated and the *static* distribution $P(q)$ of $q_{\alpha\beta}$ values displays a bimodal profile, with two *symmetric* maxima around $q = \pm 1$ [12,24]. In contrast, in the prelasing phase modes are uncorrelated and only a single maximum occurs at $q = 0$, with $P(q)$ presenting negligible values at $q = \pm 1$.

Having reviewed the equilibrium spin-glass phase that emerges from Hamiltonian (1) with fixed S, we now turn to assess its *nonequilibrium* properties, particularly in the Floquet regime, in which periodicity in time is introduced by some periodic driving source, so that $\mathcal{H}_S(t) = \mathcal{H}_S(t + T)$. As indicated from the quadratic and quartic couplings in Eq. (1), one possible experimental route to achieve this is by subjecting the RFL system to a time-dependent cw pump intensity $S(t)$, with a periodic slowly oscillating envelope of period T larger than the characteristic system times. At this point, we recall that the strong quenched disorder due to the random Bragg gratings in the erbium-based RFL system leads to spatially localized modes above threshold [25], thus providing the necessary mechanism for energy localization which is key for stabilizing the Floquet phase. Specifically, in the present RFL with $\sim 10^3$ random gratings, the average transmission spectrum $\mathrm{Tr} \sim \exp(-L/\xi)$ yields a localization length ξ that is much shorter (about six times) than the fiber length L [25].

Floquet spin-glass phases have been recently reported through theoretical studies in quantum disordered spin systems [1–7]. Notably, in these cases the periodic drive can induce two of such phases: an analog of the equilibrium spin glass phase, with the dynamics order parameter displaying the *same* period T of the drive, and a nontrivial π-spin glass with period doubling. The characterization of the Floquet spin-glass phase in magnetic systems was made, e.g., by studying the periodic oscillations of the time-dependent spin-spin correlation function.

Experimental results.—The erbium-doped RFL is a 30 cm long polarization maintaining structure [25] fabricated by CorActive (peak absorption of 28 dB/m at 1530.0 nm, mode field diameter of 5.7 μm, 0.25 numerical aperture). A spectrometer with nominal 0.2 nm resolution at 1530.0 nm was employed, with a liquid-N_2-cooled infrared CCD camera as the detector. The pump source was a semiconductor laser operating in the cw regime at 1480.0 nm. A rather long series of RFL emission spectra with intensities $\{I_\alpha(k)\}$, $\alpha = 1, 2, \ldots, N_s$, where $N_s = 150\,000$, was generated for an excitation power $P/P_{\mathrm{th}} = 2.92$ well above the RFL threshold, in which nonlinear effects are determinant, for each one of the 512 wavelengths indexed by k in the interval [1489.0 nm, 1591.4 nm] around the peak emission at 1540.0 nm (corresponding to $k = 254$).

Since an integration time $t_0 = 100$ ms takes place between two consecutive spectra acquisitions, the time variable can be defined as $t = \alpha t_0$. Two emission spectra α and β, associated with distinct RFL mode configurations, are thus separated in time by $\tau = \beta - \alpha$ (in t_0 units). In analogy to the disordered spin systems, the characterization of the photonic Floquet spin-glass phase involves the study of the time dependence of the correlation between RFL mode configurations. As argued above, in practice this can be achieved by analyzing the correlations (overlaps) between intensity fluctuations Δ_α and Δ_β associated with the spectra α and β, with $\beta = \alpha + \tau$, calculated from Eq. (2), that is, $q_\alpha(\tau) \equiv q_{\alpha,\alpha+\tau} \propto \sum_k \Delta_\alpha(k)\Delta_{\alpha+\tau}(k)$. [In the magnetic-to-photonic analogy, this corresponds to infer the time-dependent spin-spin correlator $C(t,\tau) \propto \sum_i S_i(t)S_i(t+\tau)$.] For each fixed separation time τ, a distribution $P(q) \equiv P[q(\tau)]$ of $q_\alpha(\tau)$ values is built by varying α, as displayed below.

Figure 1(a) shows the RFL emitted intensity $I_\alpha(k)$ at the peak wavelength ($k = 254$) as a function of the time

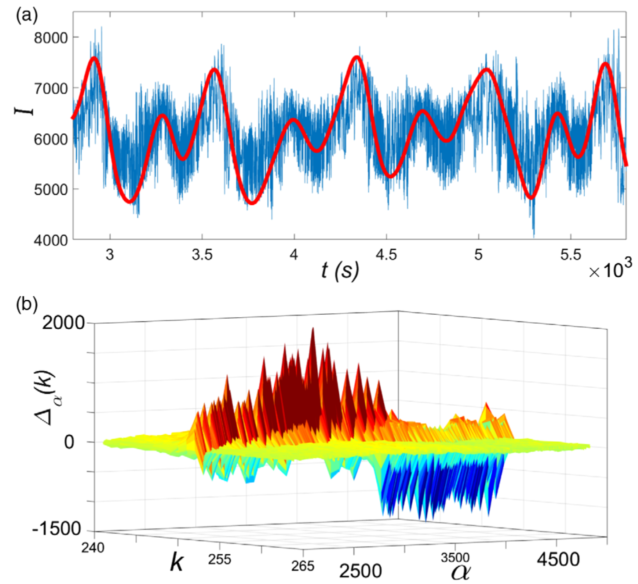

FIG. 1. (a) Time dependence of the peak intensity emitted by the erbium-based RFL pumped by a slowly varying, periodic cw driving source well above threshold (blue line). The rapid, small noisy fluctuations are superimposed by a smooth periodic low-frequency signal. The red line is an estimation of this signal using a numerical low-pass filter, with a period $T = 714$ s nearly coinciding with that of the drive. (b) Dynamics of the intensity fluctuations as function of the mode (k) and spectrum (α) labels within one period of the drive, with $t = \alpha t_0$ and $t_0 = 100$ ms.

$t = \alpha t_0$ in the range $t \in [2800\text{ s}, 5800\text{ s}]$, i.e., for α values in the interval [28 000, 58 000] out of the 150 000 RFL spectra acquired. It shows rapid, small noisy fluctuations that are superimposed by a smooth periodic low-frequency signal of period $T = 714$ s. We estimated this signal by applying a numerical low-pass filter (red line) to the raw data (blue line). We stress that the pump laser intensity also presents long-term oscillations well above threshold with essentially the same period as those of the erbium-based RFL. In order to check this property of the pump laser independently, we rearranged the experimental setup and performed a new measurement at $2P_{\mathrm{th}}$. We found that the period of the pump laser is the same (within $\pm 8\%$) as that of the RFL emitted intensity observed in Fig. 1(a). The RFL spectra can be inferred from the intensity fluctuations $\Delta_\alpha(k)$ shown in Fig. 1(b) as function of k and α, consistently with Fig. 1(a).

We now turn to investigate the dynamics of the distribution $P(q)$ of correlations between mode configurations separated in time by τ. Figure 2 shows $P(q)$ for several τ displaying a spin-glass-like bimodal profile (note the absence of the uncorrelated maximum at $q = 0$), but with a clear asymmetry observed between the maxima around $q = \pm 1$, with strong dependence on τ. The experimental results (histograms) are in nice agreement with the theoretical description (red lines, see below). The dynamical asymmetry in Fig. 2 contrasts markedly with the *undriven*

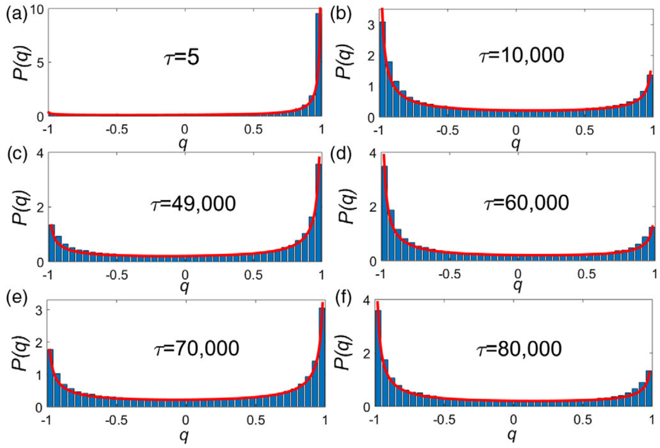

FIG. 2. Distribution $P(q)$ of correlations between mode configurations separated by a fixed time τ (histograms). A clear dynamical asymmetry between the Floquet photonic spin-glass maxima around $q = \pm 1$ is noted. In contrast, in undriven equilibrium spin-glass phases the side peaks are static and symmetric. Red lines show a nice agreement with Eq. (3).

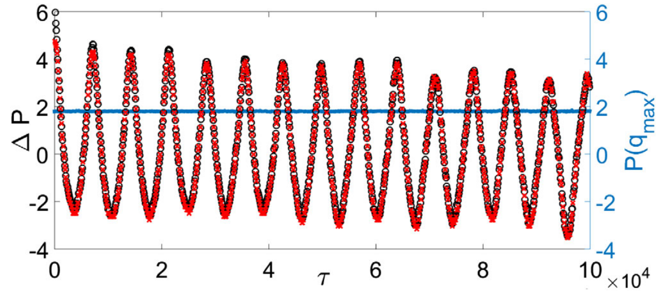

FIG. 3. Asymmetry parameter ΔP as function of the time interval τ between RFL spectra in the photonic Floquet spin-glass phase above threshold. A fine agreement between experimental (black circles) and theoretical (red circles) results is noticed. The period is the same of the drive in Fig. 1(a) (recall that $t = \alpha t_0$ and $\tau = \beta - \alpha$). In contrast, in the fluorescent regime below threshold without localization, no Floquet phase sets in, no asymmetry is found, and the maximum value of the distribution $P(q)$ (blue line) has no dynamics.

equilibrium spin-glass phase, in which the side peaks of the bimodal $P(q)$ are *symmetric* and *static*, as discussed. These findings indicate that a dynamical photonic Floquet phase with spin glass properties has set in the random lasing regime of the RFL driven by a cw pump source with same oscillation period (see below). This phase is the dynamic analog of the undriven equilibrium photonic glassy regime. The scenario changes drastically for an excitation power *below* threshold, $P/P_{\mathrm{th}} = 0.72$, in which nonlinearity is irrelevant and mode localization is prevented [29] due to the high losses in the open (cavityless) RFL system. As argued above, we observe in this case that *no* Floquet phase establishes and $P(q)$ displays a single maximum around $q = 0$, with static magnitude (see below) and negligible values at $q = \pm 1$, as typical of an uncorrelated paramagneticlike fluorescent regime.

The dynamical signature of the peak asymmetry in Fig. 2 can be quantified by the asymmetry parameter $\Delta P = P(q \to +1) - P(q \to -1)$ (in practice we consider $q = \pm 0.975$). Figure 3 shows ΔP vs τ after a detrend analysis to remove a slight tendency towards negative ΔP values. The plot of ΔP displays the same period of the drive in Fig. 1(a) (recall that $t = \alpha t_0$ and $\tau = \beta - \alpha$), and a remarkable agreement between experimental data (black circles) and the theoretical results below (red circles). Moreover, the asymmetry parameter nullifies in the fluorescent regime and the maximum value of the distribution $P(q)$ has no dynamics (blue line in Fig. 3).

The form of the asymmetric distributions $P(q)$ in Fig. 2 and asymmetry parameter ΔP in Fig. 3 can be theoretically explained as follows. The overlap parameter $q_\alpha(\tau)$, Eq. (2) with $\tau = \beta - \alpha$, is a Pearson coefficient [30] that measures the correlation between the intensity fluctuations $\Delta_\alpha(k)$ and $\Delta_\beta(k)$, with $k = 1, \ldots, N$. Because of the central limit theorem, the distribution of values q_α for fixed τ approaches a Gaussian as N increases, with a convergence rate that actually depends on the distributions of $\Delta_\alpha(k)$ and $\Delta_\beta(k)$. However, a much faster convergence to a Gaussian can be obtained [30] in terms of a new variable defined by the Fisher transform $z_\alpha(\tau) = 2\mathrm{arctanh}[q_\alpha(\tau)]$. As z_α is Gaussian distributed, with mean μ and variance σ^2, we take the inverse Fisher transform to obtain

$$P(q) = \frac{1}{\sqrt{2\pi\sigma^2}} \frac{2}{1 - q^2} \exp\left(-\frac{[2\mathrm{arctanh}(q) - \mu]^2}{2\sigma^2}\right). \qquad (3)$$

Above, μ and σ act as location and scale parameters, respectively, setting the asymmetry and shape of the distribution $P(q)$. In particular, we typically obtain bimodal (unimodal) distributions related to the photonic spin-glass (fluorescent) regime for $\sigma \gtrsim 1.5$ ($\sigma \lesssim 1.5$).

We find that all distributions in Fig. 2 are nicely described by Eq. (3), with $\sigma = 3.5$ for any τ and $\mu(\tau)$ showing an oscillatory behavior with the same period of the drive. Indeed, by plotting $\Delta P = P(q = +0.975) - P(q = -0.975)$ from Eq. (3) as red circles in Fig. 3, we observe the remarkable agreement with the experimental data (black circles), consistently with the photonic Floquet spin-glass phase dynamics set in the erbium-based RFL.

In conclusion, in this work we reported the first evidence of a dynamical Floquet spin glass phase in a laser-based photonic system with a periodic cw driving source. Disorder, eigenmode localization, and nonlinearity are key ingredients to stabilize the Floquet spin-glass phase, as indicated by the dynamics of the distribution of correlations between mode configurations and asymmetry properties, nicely described by theoretical arguments.

The experimental validation of the fundamental mechanisms theoretically proposed for stabilizing phases in driven many-body systems is hard to overemphasize.

Indeed, the Floquet mechanism is essential to understand the emergence of the periodically driven spin-glass phase in our photonic experiment. We are thus positive that our experimental observation of a photonic Floquet phase is a significant step forward in the ongoing collective effort to understand how Floquet phases emerge in actual physical systems.

We thank the financial support from the Brazilian agencies CNPq, CAPES, and FACEPE, and the Institute Nacional de Fotônica project, as well as the support from the Canada Research Chairs program of the Government of Canada.

[1] W. Berdanier, M. Kolodrubetz, S. A. Parameswaran, and R. Vasseur, Proc. Natl. Acad. Sci. U.S.A. **115**, 9491 (2018).
[2] A. Lazarides, A. Das, and R. Moessner, Phys. Rev. Lett. **115**, 030402 (2015).
[3] P. Ponte, Z. Papić, F. Huveneers, and D. A. Abanin, Phys. Rev. Lett. **114**, 140401 (2015).
[4] V. Khemani, A. Lazarides, R. Moessner, and S. L. Sondhi, Phys. Rev. Lett. **116**, 250401 (2016).
[5] D. V. Else, B. Bauer, and C. Nayak, Phys. Rev. Lett. **117**, 090402 (2016).
[6] N. Y. Yao, A. C. Potter, I. D. Potirniche, and A. Vishwanath, Phys. Rev. Lett. **118**, 030401 (2017).
[7] R. Moessner and S. L. Sondhi, Nat. Phys. **13**, 424 (2017).
[8] W. Berdanier, M. Kolodrubetz, R. Vasseur, and J. E. Moore, Phys. Rev. Lett. **118**, 260602 (2017).
[9] P. Bordia, H. Lüschen, U. Schneider, M. Knap, and I. Bloch, Nat. Phys. **13**, 460 (2017).
[10] J. Zhang, P. W. Hess, A. Kyprianidis, P. Becker, A. Lee, J. Smith, G. Pagano, I.-D. Potirniche, A. C. Potter, A. Vishwanath, N. Y. Yao, and C. Monroe, Nature (London) **543**, 217 (2017).
[11] S. Choi, J. Choi, R. Landig, G. Kucsko, H. Zhou, J. Isoya, F. Jelezko, S. Onoda, H. Sumiya, V. Khemani, C. von Keyserlingk, N. Y. Yao, E. Demler, and M. D. Lukin, Nature (London) **543**, 221 (2017).
[12] N. Ghofraniha, I. Viola, F. Di Maria, G. Barbarella, G. Gigli, L. Leuzzi, and C. Conti, Nat. Commun. **6**, 6058 (2015).
[13] S. Basak, A. Blanco, and C. López, Sci. Rep. **6**, 32134 (2016).
[14] F. Tommasi, E. Ignesti, S. Lepri, and S. Cavalieri, Sci. Rep. **6**, 37113 (2016).
[15] A. S. L. Gomes, E. P. Raposo, A. L. Moura, S. I. Fewo, P. I. R. Pincheira, V. Jerez, L. J. Q. Maia, and C. B. de Araújo, Sci. Rep. **6**, 27987 (2016).
[16] P. I. R. Pincheira, A. F. Silva, S. J. M. Carreño, A. L. Moura, S. I. Fewo, E. P. Raposo, A. S. L. Gomes, and C. B. de Araújo, Opt. Lett. **41**, 3459 (2016).
[17] A. S. L. Gomes, B. C. Lima, P. I. R. Pincheira, A. L. Moura, M. Gagné, E. P. Raposo, C. B. de Araújo, and R. Kashyap, Phys. Rev. A **94**, 011801(R) (2016).
[18] B. C. Lima, A. S. L. Gomes, P. I. R. Pincheira, A. L. Moura, M. Gagné, E. P. Raposo, C. B. de Araújo, and R. Kashyap, J. Opt. Soc. Am. B **34**, 293 (2017).
[19] E. G. Turitsyna, S. V. Smirnov, S. Sugavanam, N. Tarasov, X. Shu, S. A. Babin, E. V. Podivilov, D. V. Churkin, G. Falkovich, and S. K. Turitsyn, Nat. Photonics **7**, 783 (2013).
[20] D. V. Churkin, I. V. Kolokolov, E. V. Podivilov, I. D. Vatnik, M. A. Nikulin, S. S. Vergeles, I. S. Terekhov, V. V. Lebedev, G. Falkovich, S. A. Babin, and S. K. Turitsyn, Nat. Commun. **2**, 6214 (2015).
[21] I. R. R. González, B. C. Lima, P. I. R. Pincheira, A. A. Brum, A. M. S. Macêdo, G. L. Vasconcelos, L. d. S. Menezes, E. P. Raposo, A. S. L. Gomes, and R. Kashyap, Nat. Commun. **8**, 15731 (2017).
[22] K. S. Attri, K. Mehla, S. K. Shukla, and P. K. Singh, Sci. Rep. **8**, 2408 (2018).
[23] F. Antenucci, C. Conti, A. Crisanti, and L. Leuzzi, Phys. Rev. Lett. **114**, 043901 (2015).
[24] F. Antenucci, A. Crisanti, and L. Leuzzi, Sci. Rep. **5**, 16792 (2015).
[25] M. Gagné and R. Kashyap, Opt. Express **17**, 19067 (2009).
[26] C. Viviescas and G. Hackenbroich, Phys. Rev. A **67**, 013805 (2003).
[27] G. Hackenbroich, J. Phys. A Math. Gen. **38**, 10537 (2005).
[28] L. F. Cugliandolo, D. R. Grempel, G. Lozano, H. Lozza, and C. A. da Silva Santos, Phys. Rev. B **66**, 014444 (2002).
[29] J. Liu, P. D. Garcia, S. Ek, N. Gregersen, T. Suhr, M. Schubert, J. Mørk, S. Stobbe, and P. Lodahl, Nat. Nanotechnol. **9**, 295 (2014).
[30] H. Hotelling, J. R. Stat. Soc. Ser. B **15**, 193 (1953).